

Vinh Ly

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Summary

Software Engineer with **18 years** of experience designing and owning distributed systems at Google, YouTube, and Whatnot. End-to-end ownership of systems serving billions of daily requests. Built infrastructure in **Python**, **C++**, and **Go** across Spanner, Gmail, and live-streaming platforms. Led teams of 5-8 engineers and managed on-call rotations covering services with strict availability SLAs. Next focus: platform infrastructure and developer tooling in **Python**-centric ecosystems. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages: Python, C++, Go, SQL, TypeScript, JavaScript, HTML, CSS

Infrastructure: Distributed Systems, Linux, Kubernetes, GCP

Data: Spanner, Bigtable, Dataflow, schema migration

Reliability: on-call, SLO/SLA management, incident response, post-mortems

Leadership: team management, technical roadmaps, cross-functional collaboration, Ansible, Terraform, Prometheus, Grafana, GitHub Actions, AWS

Experience

Staff Software Engineer, YouTube | 02/2025 - Present

- Designed real-time infrastructure for live-streaming viewer features, targeting sub-100ms latency for millions of concurrent viewers across global CDN edge nodes.
- Built Python-based data pipelines processing viewer interaction events, handling 50M+ events per day with automated alerting and drift detection.

Staff Software Engineer, Whatnot | 05/2024 - 02/2025

- Architected migration of auction and payment services from Ruby monolith to Go microservices, reducing p99 latency by 35% and improving deployment frequency from weekly to daily.
- Led reliability initiative for core commerce platform serving 500K+ daily active users, establishing SLOs and reducing error budget consumption by 40%.
- Designed Python tooling for internal seller analytics dashboard, enabling real-time inventory tracking across 10K+ concurrent sellers.
- Established on-call rotation and incident response playbook for payments infrastructure processing \$10M+ in monthly transactions.

Staff Software Engineer/Manager, Google | 03/2015 - 04/2024

- Owned Spanner query engine performance for Gmail backend, implementing predictive pre-warming and adaptive GC that reduced p99 query latency by 45%.
- Managed team of 6 engineers responsible for Gmail storage infrastructure reliability, maintaining 99.99% availability for 1B+ user mailboxes.
- Designed and shipped Spanner schema migration framework supporting zero-downtime schema changes for organizational database splits and merges.
- Led migration of Gmail storage layer from custom filesystem to containerized environment, reducing infrastructure costs by 20% and enabling daily deployments.
- Defined technical roadmap for team of 8 engineers, aligning quarterly objectives with product priorities and cross-team dependencies.

Software Engineer, Google | 02/2011 - 02/2015

- Built Python library for large-scale numerical computation with C++ backends, adopted by 200+ internal teams for data processing workflows.

- Developed Spanner migration tooling for external customers, supporting billion-row table transfers with progress tracking and rollback capabilities.
- Implemented distributed caching layer for Gmail search index using Bigtable, improving cached query throughput by 3x.
- Contributed to Dataflow pipeline optimization reducing job completion times by 25% for batch processing workloads.

Software Design Engineer II, Microsoft | 10/2007 - 12/2011

- Developed Windows kernel-mode components for Office document rendering pipeline, improving file open times by 15% on large spreadsheets.
- Built C++ framework for automated regression testing of Office file format compatibility across Office versions.
- Delivered features for Office shared libraries used by 500+ internal teams, shipping under tight release deadlines.

Software Design Engineer Intern, Microsoft | 06/2006 - 08/2006

- Implemented C++ command-line utilities for internal build system configuration, reducing developer setup time for new team members.

Education

B.S. Honors degrees in Computer Science

Virginia Tech | 2004 - 2007